1) a)

$(z-(1+\mathrm{i}))(z-(3+2\mathrm{i}))$
$=z^2-(3+2\mathrm{i})z-(1+\mathrm{i})z+(1+\mathrm{i})(3+2\mathrm{i})$
$=z^2-(4+3\mathrm{i})z+1+5\mathrm{i}$

1) b)

$(z-(-3+4\mathrm{i}))(z-(1+5\mathrm{i}))$
$=z^2-(1+5\mathrm{i})z-(-3+4\mathrm{i})z+(-3+4\mathrm{i})(1+5\mathrm{i})$
$=z^2+(2-9\mathrm{i})z-23-11\mathrm{i}$

1) c)

$(z-(-2+2\mathrm{i}))(z-(2-2\mathrm{i}))$
$=z^2-(2-2\mathrm{i})z-(-2+2\mathrm{i})z+(-2+2\mathrm{i})(2-2\mathrm{i})$
$=z^2+8\mathrm{i}$

2)

$\big(z-(1+\mathrm{i})\big)\big(z-(-1+\mathrm{i})\big)\big(z-(2+2\mathrm{i})\big)$
$=(z^2-2\mathrm{i}z-2)\big(z-(2+2\mathrm{i})\big)$
$=z^3-(2+4\mathrm{i})z^2-(6-4\mathrm{i})z+(4+4\mathrm{i})$

3) a)

$(1+9\mathrm{i})^3=(1+9\mathrm{i})(1+9\mathrm{i})(1+9\mathrm{i})$
$=(1+9\mathrm{i})(-80+18\mathrm{i})$
$=-80-720\mathrm{i}+18\mathrm{i}-162$
$=-242-702\mathrm{i}$

3) b)

$(1+9\mathrm{i})^3-(5+6\mathrm{i})(1+9\mathrm{i})^2+(35+27\mathrm{i})(1+9\mathrm{i})-58-30\mathrm{i}$
$=-242-702\mathrm{i}-(5+6\mathrm{i})(-80+18\mathrm{i})-208+342\mathrm{i}-58-30\mathrm{i}$
$=-242-702\mathrm{i}+508+390\mathrm{i}-266+312\mathrm{i}$
$=0$
$\therefore z=1+9\mathrm{i}$ is a solution.

3) c)

$(z-2)\big(z-(1+9\mathrm{i})\big) = z^2-(1+9\mathrm{i})z-2z+(2+18\mathrm{i})$

$= z^2-(3+9\mathrm{i})z+(2+18\mathrm{i})$

So for some complex number k

$(z^2-(3+9\mathrm{i})z+(2+18\mathrm{i}))(z-k) = k^3-(5+6\mathrm{i})k^2+(35+27\mathrm{i})k-58-30\mathrm{i}$

Comparing constants:

$-(2+18\mathrm{i})k = -58-30\mathrm{i}$

$k = \frac{58+30\mathrm{i}}{2+18\mathrm{i}} \times \frac{2-18\mathrm{i}}{2-18\mathrm{i}} = \frac{656-984\mathrm{i}}{328} = 2-3\mathrm{i}$

Alternative Method:

Sum of roots: $5+6\mathrm{i} = (1+9\mathrm{i})+(2)+(a+b\mathrm{i})$

$\Rightarrow a=2, b=-3$ so $2-3\mathrm{i}$

4) a)

$(1+2\mathrm{i})^2-(2+3\mathrm{i})(1+2\mathrm{i})-k=0$

$-3+4\mathrm{i}+4-7\mathrm{i}-k=0$

$\therefore k = 1-3\mathrm{i}$

4) b)

$(1-2\mathrm{i})^2-(2+3\mathrm{i})(1-2\mathrm{i})-(1-3\mathrm{i})$

$= -3-4\mathrm{i}-8+\mathrm{i}-1+3\mathrm{i}$

$= -12 \neq 0$ so not a root.

5) a)

All the coefficients are not real, so must use factor theorem.

Since $2-\mathrm{i}$ is a root, $\mathrm{f}(2-\mathrm{i})=0$

$(2-\mathrm{i})^3-3\mathrm{i}(2-\mathrm{i})^2+2(2-\mathrm{i})+k=0$

$= (2-\mathrm{i})(5-4\mathrm{i})-3\mathrm{i}(5-4\mathrm{i})+4-2\mathrm{i}+k=0$

$= 2-11\mathrm{i}-12-9\mathrm{i}+4-2\mathrm{i}+k=0$

$= -6-22\mathrm{i}+k=0$

$\therefore k = 6+22\mathrm{i}$

5) b)

$(2+\mathrm{i})^3 - 3\mathrm{i}(2+\mathrm{i})^2 + 2(2+\mathrm{i}) + 6 + 22\mathrm{i}$

$= (2+\mathrm{i})(3+4\mathrm{i}) - 3\mathrm{i}(3+4\mathrm{i}) + 4 + 2\mathrm{i} + 6 + 22\mathrm{i}$

$= 2 + 11\mathrm{i} + 12 - 9\mathrm{i} + 4 + 2\mathrm{i} + 6 + 22\mathrm{i}$

$= 24 + 26\mathrm{i} \neq 0$

$\therefore$ by the factor theorem $2+\mathrm{i}$ is not a factor and so $2+\mathrm{i}$ is not a root of the equation $\mathrm{f}(z) = 0$

6) a)

$z^3 - \mathrm{i} = (z+\mathrm{i})(z^2 - \mathrm{i}z - 1)$

$\mathrm{f}(z) = z^2 - \mathrm{i}z - 1$

6) b)

$(z+\mathrm{i})(z^2 - \mathrm{i}z - 1) = 0$

Consider the quadratic factor:

$z = \frac{\mathrm{i} \pm \sqrt{(-\mathrm{i})^2 + 4}}{2} = \frac{\mathrm{i} \pm \sqrt{3}}{2} = \pm\frac{\sqrt{3}}{2} + \frac{1}{2}\mathrm{i}$

So the solutions are $z = \mathrm{i}$, $z = \frac{\sqrt{3}}{2} + \frac{1}{2}\mathrm{i}$ and $z = -\frac{\sqrt{3}}{2} + \frac{1}{2}\mathrm{i}$

7)

$z = \frac{8-5\mathrm{i} \pm \sqrt{(8-5\mathrm{i})^2 - 4(3-29\mathrm{i})}}{2}$

Consider just the discriminant:

$(8-5\mathrm{i})^2 - 4(-15\mathrm{i}) = 39 - 80\mathrm{i} - 12 + 116\mathrm{i} = 27 + 36\mathrm{i}$

So now find

$\sqrt{27+36\mathrm{i}} = a + b\mathrm{i}$ for real numbers a and b

$27 + 36\mathrm{i} = (a+b\mathrm{i})^2 = a^2 - b^2 + 2ab\mathrm{i}$

Solve

$a^2 - b^2 = 27$ and $36 = 2ab$

simultaneously:

$b = \frac{18}{a}$ so $a^2 - \left(\frac{18}{a}\right)^2 = 27$

$a^4 - 324 = 27a^2$

$a^4 - 27a^2 - 324 = 0$

$(a^2 - 36)(a^2 + 9) = 0$

Since a is real, $a^2 = 36$ so $a = 6$ or $a = -6$

So that $b = 3$ or $b = -3$

That is $\sqrt{27+36\mathrm{i}} = 6 + 3\mathrm{i}$ or $-6 - 3\mathrm{i}$

$z = \frac{8-5\mathrm{i} \pm (6+3\mathrm{i})}{2}$

$z = \frac{14}{2} - \frac{2}{2}\mathrm{i}$ or $z = \frac{2}{2} - \frac{8}{2}\mathrm{i}$

So $z = 7 - \mathrm{i}$ and $z = 1 - 4\mathrm{i}$

8)

Sum of roots: $a + 6 - 2\mathrm{i} = 10 - 2\mathrm{i}$ so $a = 4$

Product of roots: $4(6 - 2\mathrm{i}) = 24 - 8\mathrm{i}$

$\therefore k = 24 - 8\mathrm{i}$

9)

Sum of roots: $a + 2 + 7\mathrm{i} = k$

Product of roots: $a(2 + 7\mathrm{i}) = 6 + 21\mathrm{i}$

$a = \frac{6+21\mathrm{i}}{2+7\mathrm{i}} = \frac{3(2+7\mathrm{i})}{2+7\mathrm{i}} = 3$

So $3 + 2 + 7\mathrm{i} = k$

$\therefore k = 5 + 7\mathrm{i}$

10)

Since the coefficients are not real, there are no complex conjugate roots.

Sums of roots: $1 + \mathrm{i} + 3 + a = 2\mathrm{i} \therefore a = -4 + \mathrm{i}$

Sums of pairs of roots: $p = 3(1 + \mathrm{i}) + (-4 + \mathrm{i})(1 + \mathrm{i}) + 3(-4 + \mathrm{i}) = -14 + 3\mathrm{i}$

Product of roots: $q = -3(1 + \mathrm{i})(-4 + \mathrm{i}) = 15 + 9\mathrm{i}$

$\therefore p = -14 + 3\mathrm{i}$ and $q = 15 + 9\mathrm{i}$

11)

Since the coefficients are not real, there are no complex conjugate roots.

Let a be the remaining root

Sums of pairs of roots: $-4(2 + \mathrm{i}) + a(2 + \mathrm{i}) - 4a = -5 - 8\mathrm{i}$

$-8 - 4\mathrm{i} + 2a + a\mathrm{i} - 4a = -5 - 8\mathrm{i}$

$a(-2 + \mathrm{i}) = 3 - 4\mathrm{i}$

$a = \frac{3-4\mathrm{i}}{-2+\mathrm{i}} \times \frac{-2-\mathrm{i}}{-2-\mathrm{i}} = \frac{-10-5\mathrm{i}}{5} = -2 + \mathrm{i}$

Sum of roots: $p = -(-4 + (2 + \mathrm{i}) + (-2 + \mathrm{i})) = 4 - 2\mathrm{i}$

Product of roots: $q = -4(2 + \mathrm{i})(-2 + \mathrm{i}) = (-8 - 4\mathrm{i})(-2 + \mathrm{i}) = 20$

$\therefore p = 4 - 2\mathrm{i}$ and $q = 20$

12)

Since the coefficients are not real, there are no complex conjugate roots.

Let a be the remaining root

Product of roots: $\frac{13}{2} = 3(2 - 3\mathrm{i})a$

$13 = (12 - 18\mathrm{i})a$

$a = \frac{13}{12-18\mathrm{i}} \times \frac{12+18\mathrm{i}}{12+18\mathrm{i}} = \frac{156+234\mathrm{i}}{468} = \frac{1}{3} + \frac{1}{2}\mathrm{i}$

Sum of roots: $\frac{p}{2} = -\left(3 + (2 - 3\mathrm{i}) + \left(\frac{1}{3} + \frac{1}{2}\mathrm{i}\right)\right)$

$\frac{p}{2} = -\left(\frac{32}{6} - \frac{15}{6}\mathrm{i}\right) \Rightarrow p = -\frac{32}{3} + 5\mathrm{i}$

Sums of pairs of roots: $\frac{q}{2} = 3(2 - 3\mathrm{i}) + \left(\frac{1}{3} + \frac{1}{2}\mathrm{i}\right)(2 - 3\mathrm{i}) + 3\left(\frac{1}{3} + \frac{1}{2}\mathrm{i}\right)$

$\frac{q}{2} = 6 - 9\mathrm{i} + \frac{13}{6} + 1 + \frac{3}{2}\mathrm{i}$

$\frac{q}{2} = \frac{55}{6} - \frac{15}{2}\mathrm{i} \Rightarrow q = \frac{55}{3} - 15\mathrm{i}$

$\therefore p = -\frac{32}{3} + 5\mathrm{i}$ and $q = \frac{55}{3} - 15\mathrm{i}$

13) a)

Sum of roots: $10 - 2\mathrm{i} = a + b + a + b\mathrm{i}$

Considering imaginary part: $-2 = b$

Considering real part: $10 = 2a + b$

$\therefore a = 6$ and $b = -2$

13) b)

Sums of pairs of roots: $p = -2(6) - 2(6 - 2\mathrm{i}) + 6(6 - 2\mathrm{i}) = 12 - 8\mathrm{i}$

Product of roots: $-q = -2(6)(6 - 2\mathrm{i}) = -72 + 24\mathrm{i}$

$\therefore p = 12 - 8\mathrm{i}$ and $q = 72 - 24\mathrm{i}$

14)

$6 + 3\mathrm{i} = 3 + (1 + 2\mathrm{i}) + \alpha$

$6 + 3\mathrm{i} = 4 + 2\mathrm{i} + \alpha$

$\therefore 2 + \mathrm{i}$